

Yen Lin (Darren) Chao

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Education

Master of Arts in Design Engineering 2025-Present
Brown University and Rhode Island School of Design (RISD)

Bachelor of Science in Electrical Engineering and Computer Science (EECS) 2021-2024

University of California, Berkeley

Skills

Design & Fabrication

Autodesk Fusion 360, Adobe Illustrator, Laser Cutting, 3D Printing/Extrusion, Rapid Prototyping, Rhino, Arduino, ESP32, Raspberry Pi, Sheet Metal Working, Ceramic Firing, Embroidery, Wood Shop, Leather Working

Programming Languages

Python, Java, C, C++, Go, SQL, RISC-V, HTML, CSS, Javascript

Cloud & DevOps

Azure IoT Hub, Jenkins, Maven, Git, Jira, Particle Cloud, Arduino IDE, Unity

Work Experience

Co-Founder and PLA Sustainability Researcher

Re/Plastic: Brown Design Workshop

August 2025 - Present

- Optimized a closed-loop **recycling workflow**, managing the **transition of PLA waste from pelletizing and grinding to industrial extrusion and heat-pressing**.
- Architected an end-to-end **IoT monitoring system** using Particle microcontrollers and sensors to **track critical environmental variables** (humidity/temperature) during PLA processing.
- Engineered **data pipelines** by routing real-time sensor data to **Azure IoT Hub**, automating the flow into a **custom SQL database for centralized storage**.
- Conducted rigorous **material testing**, including the calibration of laser cutter settings and heat-press parameters, ensuring structural integrity of products.
- Led **physical product design initiatives**, transforming recycled PLA waste into retail-ready consumer goods sold at the Brown University Bookstore.
- Facilitated **stakeholder meetings** with retail partners and logistics teams to align fabrication capabilities with market demand and pricing strategies.

Pottery Studio Instructor

Brown Design Workshop

January 2026 - Present

- Led weekly small **group ceramics classes** with hands on instruction and individualized guidance, fostering a **supportive and collaborative learning space**.
- Managed **studio operations** including organizing kiln firings and coordinating with instructors to ensure smooth studio workflow

Software Engineer Intern

Development Bank of Singapore - Taipei

Summer 2024

- Collaborated within an **Agile team**, following **Scrum methodologies** with weekly **stand-up meetings and sprints**.
- Designed and implemented **RESTful APIs** for the "GiftBox" campaign on the CARD+ app and **manipulated XML for form structures and data**, directly interacting with the database and processing form inputs.

Design Projects

NODE

Design Engineering Robotics Kit for Middle Schoolers March 2026 - Present

- Designed and built a **modular cardboard robotics kit** integrating an **Arduino-based control system**, custom **transistor motor driver** (4 DC motors), and **plug-and-play electronics** for hands-on learning through physical building and iteration
- Rapidly **prototyped 30+ iterations** using **3D printing, laser cutting, and shop fabrication**, iterating on mechanisms, structural design, and assembly experience through fast build-test-refine cycles
- Developed full embedded **hardware system** including **LiPo power system, 5V step-up regulation, and soldered PCB** transition from breadboard prototype to compact, durable unit
- Conducted hands-on **user testing with students** to improve usability and clarity, iterating on connector design, assembly flow, and system feedback based on real-world **failure points and learning behavior**

Jukebox 2.0

Music Player that Plays Through Color

December 2025

- Designed and built an **interactive embedded system** that encourages creative music composition in children by **translating familiar color drawing inputs into real-time musical output**, mapping visual patterns into tonal sequences.
- Prototyped and iterated **Arduino-based hardware** from **breadboard to custom PCB**, improving signal stability and system integration across sensing and audio components.
- Fabricated and refined **full physical assembly** (3D-printed enclosure + paper-feed mechanism), debugging sensor noise, lighting sensitivity, and calibration to ensure reliable real-world performance.

Steady Stix

Music Player that Plays Through Color

September 2025

- Designed an **assistive dining utensil** that improves chopstick usability for **users with hand tremors**, preserving **cultural familiarity** while enabling, stigma-free eating.
- Conducted **user research, user testing**, and **biomechanical analysis** to identify anchoring opportunities in the thumb-index finger region, translating insights into functional constraints and ergonomic **design decisions**.
- Rapidly prototyped **30+ iterations within 2 weeks**, and iterated form and mechanism using **Rhino**, air-dry clay, and **3D printing**, refining geometry, balance, and hinge behavior to achieve intuitive motion and long-term durability targets.

</> Coding Projects

Idle Earth

Multi-User AI Digital Identity Simulation Platform

February 2026

- Built a **multi-user web application** where users create persistent **AI-driven “digital selves”** on a 3D earth, leaving **evolving digital artifacts** that accumulate over time.
- Developed with **Three.js** for real-time 3D globe rendering, **Google Firebase** for scalable user data storage, and **Gemini API** to simulate daily activities and generate life logs.

PACMAN

Modeling Multi-agent Game with ML

February 2024

- Utilized **Value Function**, **Q learning**, and **Deep Q Network algorithms** to enhance PACMAN's gameplay.
- Implemented **alpha-beta tree pruning** algorithm to efficiently search for and select the **optimal path** for PACMAN.

REGULUS

Secure File Sharing System

November 2023

- Crafted a **multifaceted security infrastructure** for the file sharing system, integrating robust **user verification** mechanisms to authenticate users and stringent file **modification protocols** to ensure data integrity.
- Employed cryptographic tools including **RSA** for public-key encryption, **HMACs** for message authentication, and **symmetric key encryption** for secure data transmission.
- Established secure communication channels utilizing protocols like **TLS/SSL** to protect data in transit and mitigate risks associated with interception or eavesdropping

Build Your Own World

Generating Random Explorable and Interactive Worlds

March 2023

- Architected an algorithm for generating **pseudo-random worlds** with customizable seeds in **Java**.
- Implemented a **modified Kruskal's algorithm** to connect **randomly generated rooms with hallways efficiently**.